AP Calc Lesson 2.1 Homework

Name ______________________

1. Let $f(x) = -x^2 + 7x$.

   a. Find $f(-2)$.

   b. Find $f(w)$.

   c. Find $f(x+h)$.

   d. Use your answer from part c to find an expression for $f(x+h) - f(x)$.

2. Let $f(x) = 3x^2 - 8$.

   a. Evaluate $\lim_{h \to 0} \frac{f(3+h) - f(3)}{h}$

   b. Explain what your answer in part (a) represents.

3. Lina was asked to use the limit definition of the derivative to calculate the slope of a line tangent to the graph of $f(x) = 4x - x^2$ at $x = 3$. The fist step of Lina's work is given below.

   $$\lim_{h \to 0} \frac{(12+h) - (3+h)^2 - 12 - 9}{h}$$

   a. Is Lina's work correct? If not, describe any errors she made.

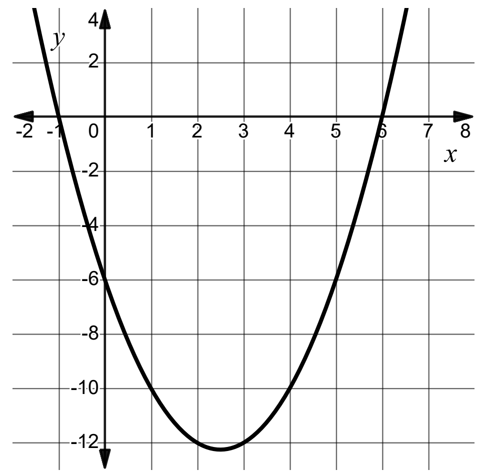
   b. Find the slope of the line tangent to f at $x = 3$.

4. Let $f(x)=x^2-5x-6$. The graph of f is shown. Evaluate $\dfrac{f(1+h)-f(1)}{h}$ for each value of h. Illustrate parts a-c on the graph of f.

a. $h=4$

b. $h=2$

c. $h=0.5$



d. Which of your values in parts a-c is closest to the instantaneous rate of change of f at $x=1$?

5. The function f is given by $f(x)=2x-\ln x$. Which of the following expressions given the instantaneous rate of change of f at $x=e$? Choose all that apply.

A. $\lim\limits_{h\to 0}\frac{((2e+h)-\ln(x+h))-(2e-1)}{h}$

B. $\lim\limits_{h\to 0}\frac{(2(e+h))-\ln(e+h))-2e-1}{h}$

C. $\lim\limits_{h\to 0}\frac{(2x-\ln x)-(2e-1)}{h}$

D. $\lim\limits_{x\to e}\frac{(2x-\ln x)-2e-1}{x-e}$

E. $\lim\limits_{x\to e}\frac{(2e+2h)-\ln(e+h)-(2e-1)}{x-e}$

F. $\lim\limits_{x\to e}\frac{(2e-1)-(2x-\ln x)}{x-e}$

G. $\lim\limits_{h\to 0}\frac{(2e+2h)-\ln(e+h)-(2e-1)}{h}$

H. $\lim\limits_{x\to e}\frac{2x-\ln x-2e+1}{x-e}$

6. The function f is given by $f(x)=4-3x$. Let $k=\dfrac{f(9)-f(7)}{9-7}$.

a. What is the value of k? How do you know ?

b. What is the value of $\lim\limits_{h\to 0}\dfrac{f(4+h)-f(4)}{h}$? How do you know?

c. What is the value of $\lim\limits_{x\to -2}\dfrac{f(x)-f(-2)}{x-(-2)}$? How do you know?

7. Let f be a function such that the slope of the line tangent to f at $x = c$ is given by $\lim_{h \to 0} \frac{\tan\left(\frac{\pi}{4} + h\right) - 1}{h}$

a. Identify the equation for $f(x)$.

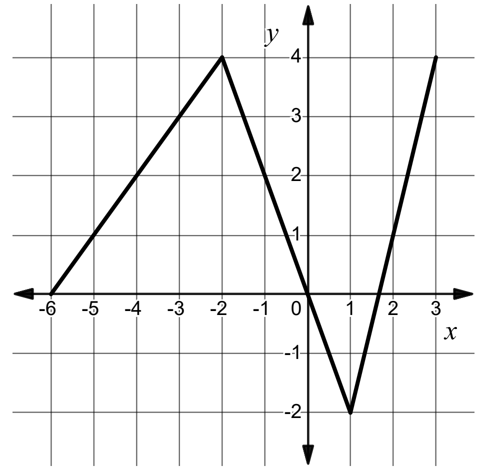
b. Find the value of c.

8. A doctor is monitoring the concentration of a medication in a patient's bloodstream. The concentration of the medication in the bloodstream, in milligrams per liter, at time t hours is given by $C(t) = \frac{20}{5+t}$.

a. Find the average rate of change of the concentration of medication in the patient's bloodstream on the interval $[1, 3]$ and label your answer with units.

b. The doctor needs to know the rate at which the concentration of the medication is changing at $t = 2$ hours. Use Definition 2 of the derivative to calculate this rate of change, in $\frac{mg/L}{hr}$.

9. The graph of the piecewise linear function $y = f(x)$ is show. Which of the following statements about f is false?



A) $\dfrac{f(-2) - f(2)}{-2 - 2} = -\dfrac{3}{4}$

B) $\lim\limits_{h \to 0} \dfrac{f(-1+h) - f(-1)}{h} = -2$

C) $\lim\limits_{x \to 2} \dfrac{f(x) - f(2)}{x - 2} = 3$

D) $\lim\limits_{x \to -4} (f(x) - f(-4)) = 1$

10. Let f be a function with $f(2) = 10,\ f(5) = 1$, and $\lim\limits_{x \to 5} \dfrac{f(x) - f(5)}{x - 5} = 1.25$. Which of the following is NOT guaranteed?

A) The average rate of change of f on $[2, 5]$ is negative.

B) The instantaneous rate of change of f at $x = 2$ is negative.

C) The slope of the secant line between $x = 2$ and $x = 5$ is -3.

D) The slope of the tangent line at $x = 5$ is positive.